

Hausübung EMF Kapitel 4:

- ① • H_1 zeigt in Uhrzeigersinn, weil der Strom des Leiters in die Zeichenebene zeigt.
- H_2 zeigt gegen Uhrzeigersinn, weil der Strom im zylindrischen Hohlraum heraus fließt.

„Rechte Handregel“

$$\|r\| \leq R \left\{ \begin{array}{l} H_1(r_1) = \frac{1}{2} \vec{j} \times \vec{r}_1 \Rightarrow \text{Spannt ein Volumen auf Stromrichtung, Radius, } H_1 \text{ mag Feld} \\ H_2(r_1) = -\frac{1}{2} \vec{j} \times \vec{r}_2 \end{array} \right. \quad \begin{array}{l} \vec{r}_2 = \vec{r}_1 - \vec{a} \\ \hookrightarrow \text{Spitze minus Schaft} \end{array}$$

$$\|r\| > R \left\{ \begin{array}{l} H_1(r_1) = \frac{R_0^2}{2 \|\vec{a}\|^2} \vec{j} \times \vec{r}_1 \\ H_2(r_1) = -\frac{r_0^2}{2 \|\vec{r}_2\|^2} \vec{j} \times \vec{r}_2 \end{array} \right.$$

für: $\|\vec{r}_1\| \leq R_0 \wedge \|\vec{r}_2\| \leq r_0$

$$\vec{H}(\vec{r}_1) = \vec{H}_1 + \vec{H}_2$$

$$\vec{H}(\vec{r}_1) = \frac{1}{2} \vec{j} \times \vec{r}_1 - \frac{1}{2} \vec{j} \times (\vec{r}_1 - \vec{a})$$

$$\vec{H}(\vec{r}_1) = \frac{1}{2} \|\vec{j}\| \cdot \|\vec{r}_1\| \cdot \sin 90^\circ - \frac{1}{2} \|\vec{j}\| \cdot \|\vec{r}_1 - \vec{a}\| \sin 90^\circ$$

$$\vec{H}(\vec{r}_1) = \frac{1}{2} \|\vec{j}\| \cdot \sin 90^\circ (\|\vec{r}_1\| - \|\vec{r}_1 - \vec{a}\|)$$

$$\vec{H}(\vec{r}_1) = \frac{1}{2} \vec{j} \times (\vec{r}_1 - (\vec{r}_1 - \vec{a}))$$

$$\underline{\underline{\vec{H}(\vec{r}_1) = \frac{1}{2} \vec{j} \times \vec{a}}}$$

für: $\|\vec{r}_1\| \leq R_0 \wedge \|\vec{r}_2\| > r_0$

$$\vec{H}(\vec{r}_1) = \vec{H}_1 + \vec{H}_2$$

$$\vec{H}(\vec{r}_1) = \frac{1}{2} \vec{j} \times \vec{r}_1 - \frac{r_0^2}{\|\vec{r}_2\|^2} \vec{j} \times \vec{r}_2$$

$$\vec{H}(\vec{r}_1) = \frac{1}{2} \|\vec{j}\| \cdot \|\vec{r}_1\| \cdot \sin 90^\circ - \frac{r_0^2}{2 \|\vec{r}_2\|^2} \|\vec{j}\| \cdot \|\vec{r}_2\| \sin 90^\circ$$

$$\vec{H}(\vec{r}_1) = \frac{1}{2} \|\vec{j}\| \cdot \sin 90^\circ \left(\|\vec{r}_1\| - \frac{r_0^2}{\|\vec{r}_2\|^2} \|\vec{r}_2\| \right)$$

$$\underline{\underline{\vec{H}(\vec{r}_1) = \frac{1}{2} \vec{j} \times \left(\vec{r}_1 - \frac{r_0^2}{\|\vec{r}_2 - \vec{a}\|^2} (\vec{r}_1 - \vec{a}) \right)}}$$



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für: $\|\vec{r}_1\| > R_0 \wedge \underbrace{\|\vec{r}_2\| > r_0}_{\text{wenn } \vec{r}_1 \text{ nicht im Hohlraum ist, ist } \vec{r}_2 > r_0}$

$$\ddot{H}(\vec{r}_1) = \ddot{H}_1 + \ddot{H}_2$$

$$\ddot{H}(\vec{r}_1) = \frac{R_0^2}{2\|\vec{r}_1\|^2} \ddot{\vec{j}} \times \vec{r}_1 - \frac{r_0^2}{2\|\vec{r}_2\|^2} \ddot{\vec{j}} \times \vec{r}_2$$

$$\ddot{H}(\vec{r}_1) = \frac{R_0^2}{2\|\vec{r}_1\|^2} \cdot \|\ddot{\vec{j}}\| \cdot \|\vec{r}_1\| \cdot \sin 90^\circ - \frac{r_0^2}{2\|\vec{r}_2\|^2} \cdot \|\ddot{\vec{j}}\| \cdot \|\vec{r}_2\| \cdot \sin 90^\circ$$

$$\ddot{H}(\vec{r}_1) = \frac{1}{2} \|\ddot{\vec{j}}\| \sin 90^\circ \cdot \left(\frac{R_0^2}{\|\vec{r}_1\|^2} \cdot \|\vec{r}_1\| - \frac{r_0^2}{\|\vec{r}_2\|^2} \cdot \|\vec{r}_2\| \right)$$

$$\underline{\underline{\ddot{H}(\vec{r}_1) = \frac{1}{2} \ddot{\vec{j}} \times \left(\frac{R_0^2}{\|\vec{r}_1\|^2} \cdot \vec{r}_1 - \frac{r_0^2}{\|\vec{r}_1 - \vec{a}\|^2} \cdot (\vec{r}_1 - \vec{a}) \right)}}$$

③ B-Feld eines Ringstromes I

Biot-Savartsches Gesetz

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \iiint \underbrace{\underbrace{\mathbf{j}(\mathbf{x})}_{\text{Stromdichte}}} \times \frac{\mathbf{r} - \mathbf{x}}{\|\mathbf{r} - \mathbf{x}\|^3} \underbrace{dV(\mathbf{x})}_{\text{Volumenelement}}$$

$$dV = A ds \quad \mathbf{j} = \frac{I}{A} \vec{e}_{ds} \Rightarrow \underbrace{\mathbf{j} dV}_{\text{Strom}} = I ds \vec{e}_{ds} = I d\vec{s}$$

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0 I}{4\pi} \int_{\mathcal{L}} \frac{d\mathbf{s} \times (\mathbf{r} - \vec{s})}{\|\mathbf{r} - \vec{s}\|^3} \cdot d\alpha$$

Um auf ds zu kommen:

$$\mathbf{s}(\alpha) = (R \cos \alpha, R \sin \alpha, 0)^T$$

↳ differenzieren

$$d\mathbf{s} = R(-\sin \alpha, \cos \alpha, 0)^T d\alpha$$

$$\mathbf{r} - \vec{s} = (x - R \cos \alpha, y - R \sin \alpha, z)^T$$

$$d\mathbf{s} \times (\mathbf{r} - \vec{s})$$

$$\begin{pmatrix} -R \sin \alpha \\ R \cos \alpha \\ 0 \end{pmatrix} \times \begin{pmatrix} x - R \cos \alpha \\ y - R \sin \alpha \\ z \end{pmatrix} = \begin{pmatrix} z R \cos \alpha \\ +z R \sin \alpha \\ -y R \sin \alpha + R^2 \sin^2 \alpha - x R \cos \alpha + R^2 \cos^2 \alpha \end{pmatrix}$$

$$= R \begin{pmatrix} z \cos \alpha \\ z \sin \alpha \\ -y \sin \alpha + R \sin^2 \alpha - x \cos \alpha + R \cos^2 \alpha \end{pmatrix} = R \begin{pmatrix} z \cos \alpha \\ z \sin \alpha \\ \underbrace{R(\sin^2 \alpha + \cos^2 \alpha)}_1 - x \cos \alpha - y \sin \alpha \end{pmatrix}$$



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$$B(r) = \frac{\mu_0 I}{4\pi} \int_L \frac{ds \times (r - \vec{s})}{\|r - \vec{s}\|^3} dL$$

$$B(r) = \frac{\mu_0 I R}{4\pi} \int_L \frac{(z \cos \lambda, z \sin \lambda, R - x \cos \lambda - y \sin \lambda)}{\sqrt{(x - R \cos \lambda)^2 + (y - R \sin \lambda)^2 + z^2}^3} d\lambda$$

$$\begin{aligned} \hookrightarrow NR: & x^2 - 2xR \cos \lambda + R^2 \cos^2 \lambda + y^2 - 2yR \sin \lambda + R^2 \sin^2 \lambda + z^2 \\ & \underbrace{x^2 + y^2 + z^2}_{r^2} + \underbrace{R^2 (\cos^2 \lambda + \sin^2 \lambda)}_{R^2} + 2R(-x \cos \lambda - y \sin \lambda) \\ & r^2 + R^2 - 2R(x \cos \lambda + y \sin \lambda) \end{aligned}$$

$$B(x, y, z) = \frac{\mu_0 I R}{4\pi} \int_L \frac{(z \cos \lambda, z \sin \lambda, R - x \cos \lambda - y \sin \lambda)}{\sqrt{r^2 + R^2 - 2R(x \cos \lambda + y \sin \lambda)}^3} d\lambda$$

$$B(0, 0, z) = \frac{\mu_0 I R}{4\pi} \int_{-2\pi}^{2\pi} \frac{(z \cos \lambda, z \sin \lambda, R)}{\sqrt{z^2 + R^2}^3} d\lambda \quad \begin{array}{l} r^2 = x^2 + y^2 + z^2 \\ x=0 \quad y=0 \end{array}$$

$$B(0, 0, z) = \frac{\mu_0 I R}{4\pi \sqrt{z^2 + R^2}^3} \int_{-2\pi}^{2\pi} (z \cos \lambda, z \sin \lambda, R) d\lambda$$

$$B(0, 0, z) = \frac{\mu_0 I R}{4\pi \sqrt{z^2 + R^2}^3} \left(\begin{bmatrix} z \sin 2\pi, -z \cos 2\pi, R 2\pi \end{bmatrix} - \begin{bmatrix} z \sin 0, -z \cos 0, 0 \end{bmatrix} \right)$$

$$B(0, 0, z) = \frac{\mu_0 I R}{4\pi \sqrt{z^2 + R^2}^3} \cdot \begin{pmatrix} 0 \\ 0 \\ 2\pi R \end{pmatrix} = \underline{\underline{\frac{\mu_0 I R^2}{2 \sqrt{z^2 + R^2}}}}$$

B_z wird maximal bei $z=0$

$$B_z(0, 0, 0) = \frac{\mu_0 I}{2R}$$



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- ④ Helmholtz spule: ↗ muss zweimal Abgeleitet werden um den Sattelpunkt zu finden.

$$B_z(z) = \frac{\mu_0 N I R^2}{2} \left[\frac{1}{\left(\left(\frac{h}{2} + z\right)^2 + R^2\right)^{3/2}} + \frac{1}{\left(\left(-\frac{h}{2} + z\right)^2 + R^2\right)^{3/2}} \right]$$

$$B_z(z) = \frac{\mu_0 N I R^2}{2} \left[\left(\frac{h^2}{4} + hz + z^2 + R^2\right)^{-3/2} + \left(z^2 - zh + \frac{h^2}{4} + R^2\right)^{-3/2} \right]$$

$$\frac{dB_z(z)}{dz} = \frac{\mu_0 N I R^2}{2} \left[-\frac{3}{2} \left(\frac{h^2}{4} + hz + z^2 + R^2\right)^{-5/2} \cdot (h + 2z) - \frac{3}{2} \left(z^2 - zh + \frac{h^2}{4} + R^2\right)^{-5/2} \cdot (2z - h) \right]$$

$$\frac{dB_z(z)}{dz} = -\frac{\mu_0 N I R^2 3}{2} \left(\frac{\left(\frac{h}{2} + z\right)}{\left(\frac{h^2}{4} + hz + z^2 + R^2\right)^{5/2}} + \frac{\left(z - \frac{h}{2}\right)}{\left(z^2 - zh + \frac{h^2}{4} + R^2\right)^{5/2}} \right)$$

$$\frac{d^2 B_z(z)}{dz^2} = -\frac{\mu_0 N I R^2 3}{2} \left(\frac{\left(\frac{h^2}{4} + hz + z^2 + R^2\right)^{5/2} - \left(\frac{h}{2} + z\right) \cdot \frac{5}{2} \left(\frac{h^2}{4} + hz + z^2 + R^2\right)^{3/2} \cdot (h + 2z)}{\left(\frac{h^2}{4} + hz + z^2 + R^2\right)^{10/2}} + \frac{\left(z^2 - zh + \frac{h^2}{4} + R^2\right)^{5/2} - \left(z - \frac{h}{2}\right) \cdot \frac{5}{2} \left(z^2 - zh + \frac{h^2}{4} + R^2\right)^{3/2} \cdot (2z - h)}{\left(z^2 - zh + \frac{h^2}{4} + R^2\right)^{10/2}} \right)$$



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④

$$\frac{d^2 B(z)}{dz^2} = - \frac{\mu_0 N I R^2}{2} \left(\frac{1}{\left(\frac{h^2}{4} + hz + z^2 + R^2\right)^{\frac{5}{2}}} - \frac{5\left(\frac{h}{2} + z\right)^2}{\left(\frac{h^2}{4} + hz + z^2 + R^2\right)^{\frac{7}{2}}} + \frac{1}{\left(z^2 - zh + \frac{h^2}{4} + R^2\right)^{\frac{5}{2}}} - \frac{5\left(z - \frac{h}{2}\right)^2}{\left(z^2 - zh + \frac{h^2}{4} + R^2\right)^{\frac{7}{2}}} \right)$$

$$\frac{d^2 B(z)}{dz^2} = - \frac{3\mu_0 N I R^2}{2} \left(\frac{1}{\left(\left(\frac{h}{2} + z\right)^2 + R^2\right)^{\frac{5}{2}}} - \frac{5\left(\frac{h}{2} + z\right)^2}{\left(\left(\frac{h}{2} + z\right)^2 + R^2\right)^{\frac{7}{2}}} + \frac{1}{\left(\left(z - \frac{h}{2}\right)^2 + R^2\right)^{\frac{5}{2}}} - \frac{5\left(z - \frac{h}{2}\right)^2}{\left(\left(z - \frac{h}{2}\right)^2 + R^2\right)^{\frac{7}{2}}} \right)$$

$$\frac{dB(z)}{dz} = 0 \quad \text{für } z=0$$

$$\frac{d^2 B_z}{dz^2} \Big|_{z=0} = 0 = - \frac{3\mu_0 N I R^2}{2} \left(\frac{1}{\left(\frac{h^2}{4} + R^2\right)^{\frac{5}{2}}} - \frac{5\frac{h^2}{4}}{\left(\frac{h^2}{4} + R^2\right)^{\frac{7}{2}}} + \frac{1}{\left(\frac{h^2}{4} + R^2\right)^{\frac{5}{2}}} - \frac{5\frac{h^2}{4}}{\left(\frac{h^2}{4} + R^2\right)^{\frac{7}{2}}} \right)$$

$$0 = \frac{2}{\left(\frac{h^2}{4} + R^2\right)^{\frac{5}{2}}} - \frac{10\frac{h^2}{4}}{\left(\frac{h^2}{4} + R^2\right)^{\frac{7}{2}}} \quad | \cdot \left(\frac{h^2}{4} + R^2\right)^{\frac{7}{2}} | : 2$$

$$0 = 1 - \frac{5\frac{h^2}{4}}{\frac{h^2}{4} + R^2} \quad | \cdot \frac{h^2}{4} + R^2$$

$$0 = \frac{h^2}{4} + R^2 - 5\frac{h^2}{4}$$

$$4\frac{h^2}{4} = R^2 \Rightarrow 4h^2 = 4R^2 \quad | : 4 \sqrt{\quad} \Rightarrow \underline{h = R}$$

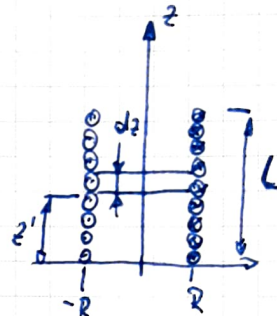


(5)

Feld einer kreisförmigen Zylindrischen Spule:

$$B_z(0,0,z) = \frac{\mu_0 I}{2} \frac{R^2}{\sqrt{z^2 + R^2}^3}$$

$$dB_z(z') = \frac{\mu_0 dI}{2} \frac{R^2}{\sqrt{(z-z')^2 + R^2}^3}$$



Gesamt durchflutung $\Rightarrow \Phi = NI$ ^{Wicklungsanzahl}

Strom pro Längeneinheit $\frac{NI}{L}$: $dI = \frac{NI}{L} dz$

Integrationsgrenzen $z'=0 - z'=L$ erhält man ges. Flussdichte

$$B_z(z) = \int_{z'=0}^L \frac{\mu_0 NI}{2L} \cdot \frac{R^2}{\sqrt{(z-z')^2 + R^2}^3} dz'$$

$$(z-z')^2 = (z'-z)^2$$

$\hookrightarrow$ Integral nicht negativ

$$B_z(z) = \int_{z'=0}^L \frac{\mu_0 NI}{2L} \cdot \frac{R^2}{R^3} \cdot \frac{1}{\sqrt{1 + \left(\frac{z'-z}{R}\right)^2}^3} dz'$$

substitution $s = \frac{z'-z}{R}$

$$dz' = R ds$$

$$\frac{ds}{dz'} = \frac{1}{R}$$

$$R ds = dz'$$